

22AIE303 DBMS Phase 3 – Normalization

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Insert Anomalies:

1. To add a new person's details, you must provide unrelated service request information.
2. To add a new service type, you must include unrelated person or staff details.
3. If certain fields like request_id or staff_id are unknown, they must be left null.

Delete Anomalies:

1. Deleting all service requests for a person might cause the person's details to be lost.
2. Removing all service requests for a staff member could result in losing staff information.

Update Anomalies:

1. Changing a person's department requires updating multiple rows.
2. Updating a staff member's specialization needs changes across multiple service request rows.
3. Modifying service type details requires updates across multiple request entries.

Normalization:

```
ICTS_Universal (  
  person_id, name, phone_no, department,  
  approval_required,  
  request_id, request_date, status, description,  
  service_type_id, service_type_name, service_category,  
  staff_id, staff_name, staff_phone, staff_specialization,  
  hod_id, hod_name,  
  approval_status  
)
```

Functional Dependencies:

1. $\text{person_id} \rightarrow \text{name, phone_no, department, approval_required}$
2. $\text{request_id} \rightarrow \text{request_date, status, description}$
3. $\text{service_type_id} \rightarrow \text{service_type_name, service_category}$
4. $\text{staff_id} \rightarrow \text{staff_name, staff_phone, staff_specialization}$
5. $\text{hod_id} \rightarrow \text{hod_name}$
6. $\text{request_id} \rightarrow \text{staff_id, hod_id, approval_status}$

Normalization to 1NF (First Normal Form):

Person (person_id, name, phone_no, department, approval_required)

Person_Contact (person_id, contact_number)

Service_Request (request_id, person_id, service_type_id, request_date, status, description)

Service_Type (service_type_id, service_type_name, service_category)

Staff (staff_id, staff_name, staff_phone, staff_specialization)

Staff_Contact (staff_id, contact_number)

HOD (hod_id, hod_name, department_id)

Request_Handling (request_id, staff_id, hod_id, approval_status)

Normalization to 2NF (Second Normal Form):

Person (person_id, name, phone_no, department, approval_required)

Person_Contact (person_id, contact_number)

Service_Request (request_id, person_id, service_type_id, request_date, status, description)

Service_Type (service_type_id, service_type_name, service_category)

Staff (staff_id, staff_name, staff_phone, staff_specialization)

Staff_Contact (staff_id, contact_number)

HOD (hod_id, hod_name, department_id)

Request_Handling (request_id, staff_id, hod_id, approval_status)

Normalization to 3NF (Third Normal Form):

Person (person_id, name, department_id, approval_required)

Person_Contact (person_id, contact_number)

Department (department_id, department_name)

Service_Request (request_id, person_id, service_type_id, request_date, status, description)

Service_Type (service_type_id, service_type_name, service_category)

Staff (staff_id, staff_name, staff_department_id, staff_specialization)

Staff_Contact (staff_id, contact_number)

HOD (hod_id, hod_name, department_id)

Request_Handling (request_id, staff_id, hod_id, approval_status)